Air University - Islamabad
Islamabad, Pakistan

Lab Task 03

Full Stack Programming Lab (B)

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Lab Tasks

Github: https://github.com/231966-ahmed/Full-stack-programming-lab-231966

Lab Task 1 - Student Management System:

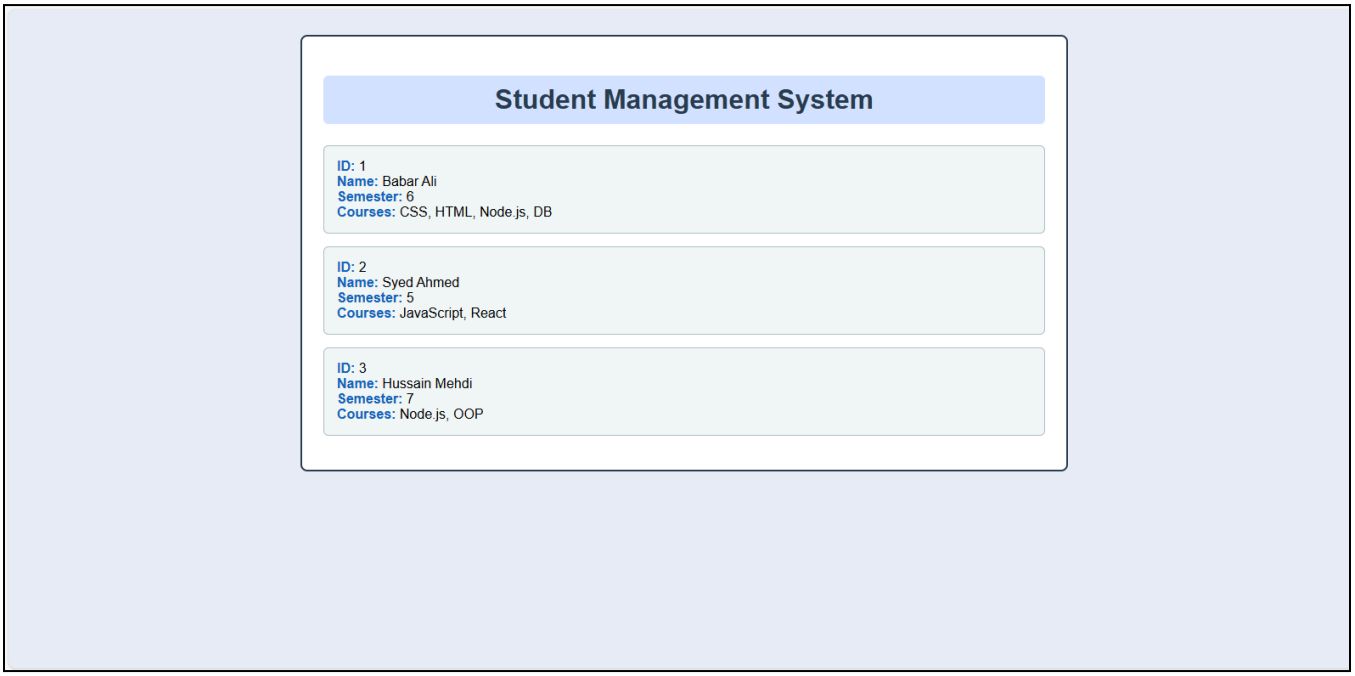
Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Management System</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            background-color: #e9eef5;
            padding: 30px;
        }

        .container {
            max-width: 850px;
            margin: auto;
            background-color: #ffffff;
            padding: 25px;
            border-radius: 8px;
            border: 2px solid #2c3e50;
        }

        h1 {
            text-align: center;
            margin-bottom: 25px;
            color: #2c3e50;
            background-color: #d6e4ff;
            padding: 10px;
            border-radius: 6px;
        }

        .card {
            background-color: #f4f6f9;
            padding: 15px;
```

```
            margin-bottom: 15px;
            border-radius: 6px;
            border: 1px solid #b0bec5;
        }

        .card:hover {
            background-color: #e3f2fd;
        }

        .card b {
            color: #1565c0;
        }
    </style>
</head>

<body>
    <div class="container">
        <h1>Student Management System</h1>
        <div id="output"></div>
    </div>
    <script>
        class Student {
            constructor(id, name, semester, courses) {
                this.id = id;
                this.name = name;
                this.semester = semester;
                this.courses = courses;
            }
            getDetails() {
                return `
                    <div class="card">
                    <b>ID:</b> ${this.id}<br>
                    <b>Name:</b> ${this.name}<br>
                    <b>Semester:</b> ${this.semester}<br>
                    <b>Courses:</b> ${this.courses.join(", ")}
                    </div>`;
            }
        }
        const students = [
            new Student(1, "Babar Ali", 6, ["CSS", "HTML", "Node.js", "DB"]),
            new Student(2, "Syed Ahmed", 5, ["JavaScript", "React"]),
```

```
            new Student(3, "Hussain Mehdi", 7, ["Node.js", "OOP"])
        ];
        let result = "";
        students.forEach(s => result += s.getDetails());
        document.getElementById("output").innerHTML = result;

    </script>
</body>
</html>
```

Output:



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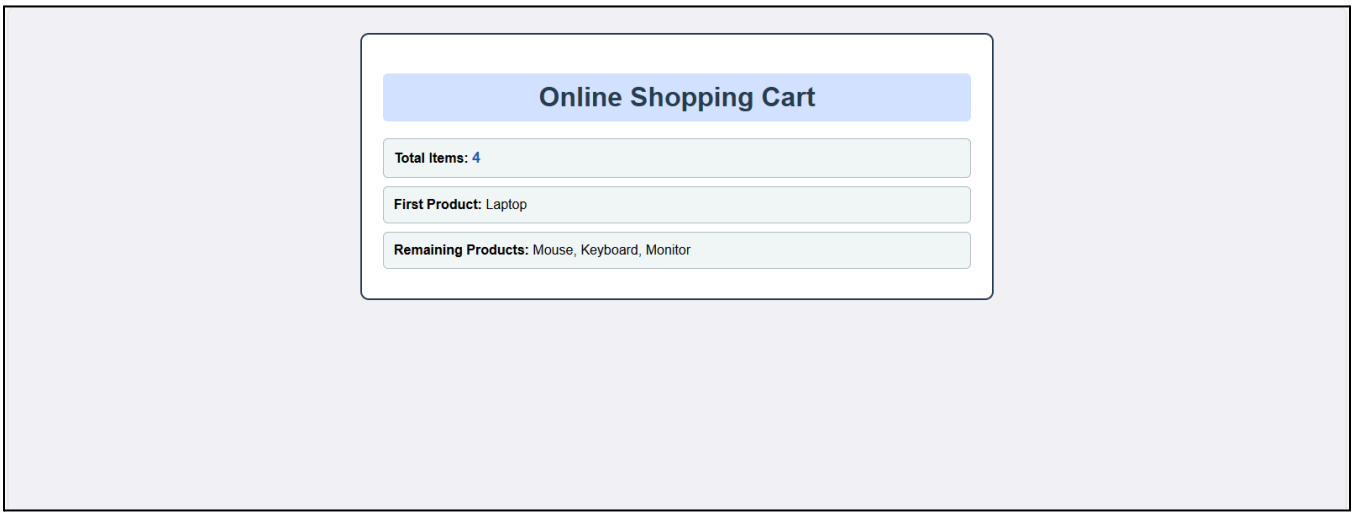
Lab Task 2 - Online Shopping Cart:

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Shopping Cart</title>
    <style>
        body
        {
            font-family: Arial, sans-serif;
            margin: 0;
            background-color: #f0f4f8;
            padding: 30px;
        }

        .box
        {
            max-width: 700px;
            margin: auto;
            background-color: #ffffff;
            padding: 25px;
            border-radius: 10px;
            border: 2px solid #2c3e50;
        }

        h1
        {
            text-align: center;
            margin-bottom: 20px;
            color: #2c3e50;
            background-color: #d6e4ff;
            padding: 10px;
            border-radius: 6px;
        }

        .info
        {
            background-color: #f4f6f9;
            padding: 12px;
            border-radius: 6px;
```

```
            margin: 10px 0;
            border: 1px solid #b0bec5;
        }

        .info:hover
        {
            background-color: #e3f2fd;
        }

        .highlight
        {
            font-size: 18px;
            color: #1565c0;
            font-weight: bold;
        }
    </style>
</head>

<body>
    <div class="box">
        <h1>Online Shopping Cart</h1>
        <div id="output"></div>
    </div>

    <script>
        function addToCart(...items) {return items;}

        const products=["Laptop", "Mouse", "Keyboard", "Monitor"];
        let cart=addToCart(...products);
        const clonedCart=[...cart];
        const[firstProduct, ...remainingProducts]=clonedCart;
        document.getElementById("output").innerHTML = `
            <div class="info"><b>Total Items:</b> <span
class="highlight">${clonedCart.length}</span></div>
            <div class="info"><b>First Product:</b> ${firstProduct}</div>
            <div class="info"><b>Remaining Products:</b>
${remainingProducts.join(", ")}</div>
            `;
    </script>
</body>
</html>
```

Output:



Lab Task 3 - Asynchronous Data Loader:

Code:

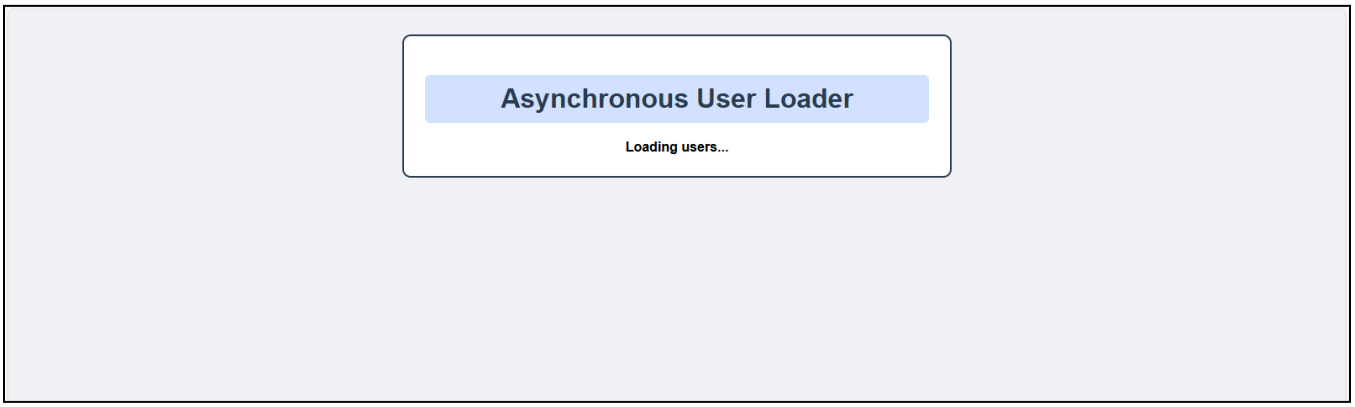
```
<!DOCTYPE html>
<html>
<head>
    <title>Async Data Loader</title>
    <style>
        body
        {
            font-family: Arial, sans-serif;
            margin: 0;
            background-color: #f0f4f8;
            padding: 30px;
        }

        .container
```

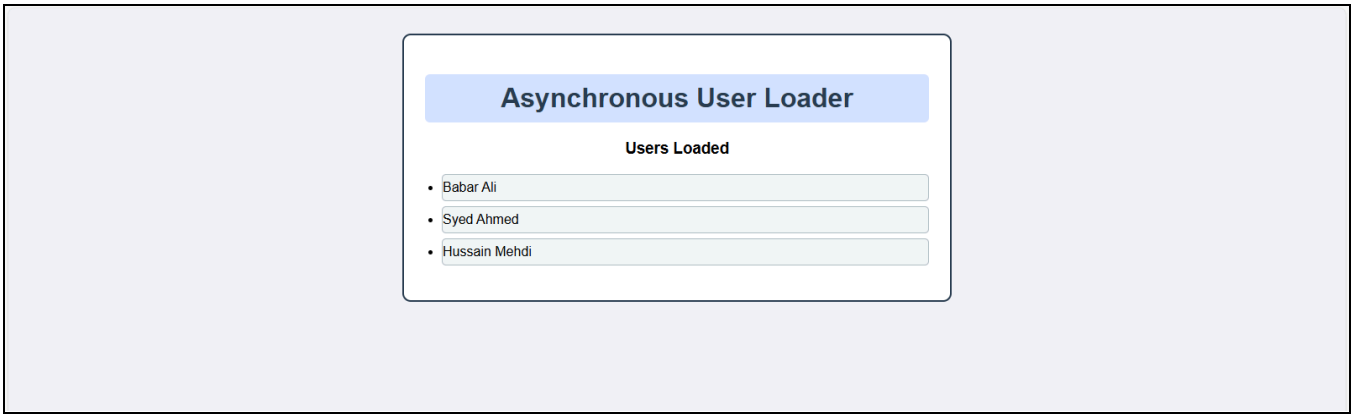
```
{
    max-width: 600px;
    margin: auto;
    background-color: #ffffff;
    padding: 25px;
    border-radius: 10px;
    text-align: center;
    border: 2px solid #2c3e50;
}

h1
{
    margin-bottom: 20px;
    color: #2c3e50;
    background-color: #d6e4ff;
    padding: 10px;
    border-radius: 6px;
}

#output
{
    margin-top: 20px;
}

ul
{
```

```
            text-align: left;
            padding-left: 20px;
        }

        li
        {
            padding: 6px 0;
            background-color: #f4f6f9;
            margin-bottom: 6px;
            border-radius: 5px;
            border: 1px solid #b0bec5;
        }

        li:hover
        {
            background-color: #e3f2fd;
        }
    </style>
</head>

<body>
    <div class="container">
        <h1>Asynchronous User Loader</h1>
        <div id="output"><b>Loading users...</b></div>
    </div>
    <script>
```

```
        function fetchUsers()
        {
            return new Promise((resolve, reject) => {
                const success = true;
                setTimeout(() => {
                    if (success) {
                        resolve([{ name: "Babar Ali" }, { name: "Syed Ahmed" }, { name: "Hussain Mehdi" }]);
                    } else {
                        reject("Error loading users");
                    }
                }, 3000);
            });
        }

        fetchUsers()
            .then(users => {
                let result = "<h3>Users Loaded</h3><ul>"; users.forEach(u => result += `<li>${u.name}</li>`); result += "</ul>"; document.getElementById("output").innerHTML = result;
            })
            .catch(e => {
                document.getElementById("output").innerHTML = `<b>${e}</b>`;
            });
    </script>
</body>
</html>
```

Output:


Asynchronous User Loader
Loading users...


Asynchronous User Loader
Users Loaded
Babar Ali
Syed Ahmed
Hussain Mehdi